Song-Ze (Jimmy) Yu 游松澤

E-mail: vaclis@berkeley.edu | **Personal Page:** https://vaclis.net/ | **GitHub:** github.com/vaclisinc

Research Interests

My research focuses on grounded **music understanding** in large audio-language models and **controllability for musicians** in music and audio effects generation. My long-term goal is to bring music models toward **artificial music intelligence**, offering musicians new interfaces to directly discuss, collaborate, and interact with machines.

Education

University of California, Berkeley Aug. 2025 - May 2026
Visiting Student in Computer Science
- GPA: 3.9 / 4.0
- Selected Coursework: Machine Learning, LLM Post-Training & AI Agents, Deep Reinforcement Learning, Computer Vision, Computer Programming for Music Applications *(CS 189, CDSS 94, CS 185, CS 180, MUSIC 159)*.

National Tsing Hua University Sep. 2023 - Jun. 2026
B.S. in Computer Science
- GPA: 3.91 / 4.30

National Cheng Kung University Sep. 2022 - Jun. 2023
B.S. in Computer Science and Information Engineering
- GPA: 4.16 / 4.30; Rank: **5 / 113** (Top 5%)

Publications (* indicates equal contribution)

Music Understanding in Audio-Language Models

[1] PitchBench: Measuring Pitch Hearing in Audio-Language Models — **NeurIPS 2026** Evaluations & Datasets
Milan Liessens Dujardin*, **Song-Ze Yu***, Craver Corbyn Thomas-Smith, David M. Chan, Karina Nguyen

[2] MuSP-Bench: Advanced Multimodal Benchmarking of Music Understanding across Score and Performance — **ISMIR 2026** Late-Breaking Demo
Milan Liessens Dujardin, **Song-Ze Yu**, Kevin Miao

[3] UniFront: Revisiting Time-Frequency Frontends for Unified Music and Speech Modeling
Song-Ze Yu, David M. Chan

[4] Diagnosing Music Understanding in Audio-Language Models: Encoder-Decoder Probing and Task Adaptation
Jun-You Wang, **Song-Ze Yu**, Xuanjun Chen, Chih-Kai Yang, Ke-Han Lu, Sung-Feng Huang

[5] Unifying Score and Performance for Fine-Grained Music Understanding in Audio-Language Models
Milan Liessens Dujardin, **Song-Ze Yu**, Kevin Miao

Controllability for musicians

[6] InstructFX2FX: A Multi-Turn Text-to-Effect System for Sequential Audio Effect Refinement — **DAFx 2026** Demonstration
Song-Ze Yu, Milan Liessens Dujardin, Yuxuan Cai, Wantong Zhang, Brian Cruz, Jeremy Wagner, Carmine-Emanuele Cella

[7] MIDI-REPA: Improving Text-to-Music Generation with Symbolic Representation Alignment
Song-Ze Yu, Fang-Duo Tsai, Yi-Hsuan Yang

HCI & human simulation

[8] Anamnesis: An Open-Source Platform for Large-Scale Backstory-Conditioned Survey Simulation — **EMNLP 2026** System Demonstration
Song-Ze Yu, Joseph Suh, Serina Chang, David M. Chan

Research Experience

Music and AI Lab, National Taiwan University Jun. 2026 - Present
Research Assistant *Advised by Prof. Yi-Hsuan Yang*
- Leading the development of a **large audio-language model (LALM) for music**, bringing together insights from prior music benchmarking [1,2], methodology [3,5], and diagnostic studies [4] into large-scale model alignment and training.
- Researching symbolic representation alignment for text-to-music generation and representation transfer and utilization in audio-language models, leading to **MIDI-REPA** [7] and **encoder–decoder probing studies** [4].

Berkeley AI Research (BAIR) Lab, UC Berkeley Feb. 2026 - Present
Researcher *Advised by David M. Chan and Prof. Trevor Darrell*
- Revisiting the ubiquitous 25 ms log-Mel frontend used across general audio encoders and models under conflicting speech–music resolution demands, leading to **UniFront** [3].
- Lead-developed **PitchBench** [1] and **Anamnesis** [8], spanning the basics of music understanding in audio-language models and large-scale human simulation.

Center for New Music and Audio Technologies Lab, UC Berkeley Sep. 2025 - Jun. 2026
Undergraduate Research Lead *Advised by Prof. Carmine-Emanuele Cella*

- Led the research direction on interactive, musician-centered audio effect modeling and editing, extending single-shot methods toward iterative refinement and leading to **InstructFX2FX** [6].

Acoustics and Hearing Group, National Tsing Hua University Jul. 2024 - May 2025
Undergraduate Researcher *Advised by Prof. Yi-Wen Liu*

- Developed the **audio-to-equalizer model (VTR)** for single-shot tone replication as my undergraduate capstone project, later deployed as a Virtual Studio Technology (VST) plugin at Positive Grid.

Selected Engineering Experience

BerkeleyTime *Berkeley, CA*

Machine Learning Pod — *Pod Lead* Jan. 2026 - May 2026

- Led a **10-member machine learning team** across semantic search, recommendation system, and enrollment prediction, coordinating researcher–engineer pairs and designers from model development through deployment.
- Led development of a **new Explore page**, integrating course recommendations into BerkeleyTime's product experience.

Catalog Pod — *Machine Learning Engineer* Sep. 2025 - Dec. 2025

- Built **BGE text-embedding and FAISS vector-search services** with FastAPI for **122K+ monthly users**.
- Reduced GraphQL first-load latency from **25s to 1s** with pagination and Redis caching; shipped and fixed **20+ pull requests**.

Positive Grid *Taipei, Taiwan*

Machine Learning Intern Jun. 2025 - Aug. 2025

- Served as Scrum Master for **Amp-AI-SaaS** and designed the **Multi-Agent Orchestration Harness** for ongoing software-engineering workflows at Positive Grid.
- Integrated my capstone **VTR audio-to-EQ model** into a **JUCE VST plugin**, using PyTorch inference for real-time audio processing in a digital audio workstation.

Selected Music Experience

Pianist, Composer & Producer Music CV | YouTube (7K subscribers)

- Performed **24 solo piano recitals** across Greater China and appeared on **70+ TV shows and New Year galas (春晚)** between 2009 and 2016.
- Shared the stage with **Jay Chou (周杰倫) three times** and performed eight-hands piano at Taipei Arena with Bai-An (白安) and Cosmos People (宇宙人) as a guest at Yan Jue's (嚴爵) concert.
- Wrote **20+ original songs with 20K+ views** and continues to compose, perform, and produce original music.

Awards & Invited Talks

Meichu Hackathon — 3rd Place, NXP Group Sep. 2026

- Built GodEyes, a **physical AI** system using a **world model** to reconstruct crime scenes as explorable 3D worlds, with SLAM-based navigation and on-device NPU gesture tracking on an NXP i.MX 93 edge board.

Claude × a16z × Berkeley M.E.T. Makeathon — Best Design Oct. 2025

- Built Haven in five hours, a mobile app that personalizes home-design recommendations from room scans.

Meichu Hackathon — 2nd Place, Google Group Oct. 2024

- Built Travelity, an AI travel planner using interests, personality, and real-time map and search data.

Taipei Hackathon — 3rd Place Sep. 2024

- Built ParkFlow, a WebView microservice for real-time parking information, navigation, and notifications; selected for integration into Taipei's official city app (**3M+ downloads**) and awarded NT$100K.
- Documented the project's development in a video reaching **1K+ views**; delivered an invited talk hosted by the Taipei City Government the following year.

SITCON Hackathon — Group Award Winner Jul. 2024

- Built ccDiary, an AI-assisted mental-health journal with emotion analysis and anonymous social features.

National Cheng Kung University — Campus Speaker Aug. 2023

- Delivered an invited talk for incoming students on writing music as a diary, then turned audience-submitted phrases and sentences on Slido into improvised songs performed live.